



CJC

Complete Java Classes

by Kunal Sir

What is java exception?

The **unwanted situation and abnormal condition** which will disturb the normal flow of program that is called as exception in java.

What is exception handling in java?

The unwanted situation and abnormal condition which will disturb the normal flow of program or sometimes it will terminate the flow of program so we can handle this situation by the help of try and catch block that is called as exception handling in java.

To better understand exceptions and exception handling, let's make a real-time comparison-

Imagine that we order a product online, but while en-route, there's a failure in delivery. A good company can handle this and gracefully re-route our package so that it still arrives on time.



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Likewise, in java, the code can experience errors while executing our instructions. Good exception handling can handle error and gracefully re-route the program to give the user still a positive experience.

Important Keyword in Exception Handling:

Try, Catch, Finally, Throws and Throw.

Major Reason Why an Exception Occurs: -

- Invalid user input
- Device failure
- Loss of network connection
- Coding errors
- Opening an unavailable file.

Exception and Error Hierarchy Chart:

There are different types of exception and errors in java which is given in below chart-



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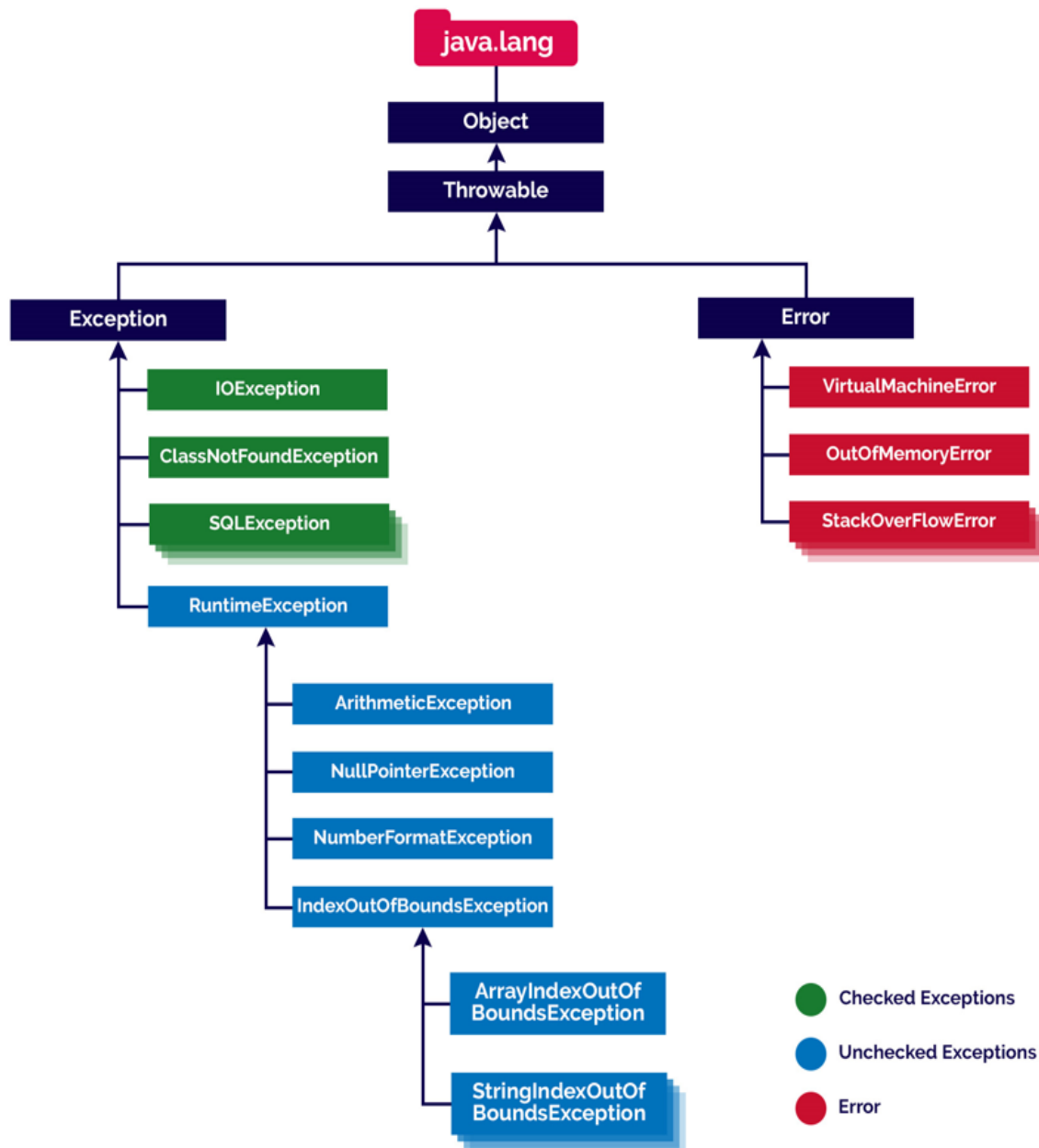


diagram of exception & error



What is an Error?

Error did not need to handle but it should be solved.

Error is usually caused by serious problems that are outside the control of the program, such as running out of memory or a system crash.

Error is represented by the error class and its sub-classes.

What is try (block and keyword) in java?

- The **try** block contains a set of statements where an exception can occur.
- In other words, we can say that **try** block is the collection of doubt statements.

Syntax: -

```
try
{

    // statements that might cause exception

}
```



What is catch (block and keyword) in java?

- The **catch** block is used to handle the uncertain conditions of a try block.
- A try block is always followed by a catch block, which handles the exception that occurs in the associated try block.
- We can able to write single try with multiple catch block.

Syntax: -

```
catch  
{  
    // statements that handle an exception  
}
```

What is the difference between Error and Exception?

There are following difference between error and exception which is given below-



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S.No.	Error	Exception
1.	Errors primarily arise due to the lack of system resources.	Exceptions may occur during both runtime and compile time.
2.	Recovery from an error is typically not possible.	Recovery from an exception is possible.
3.	All errors in java are unchecked.	Exceptions in java can be either checked or unchecked.
4.	The system executing the program is responsible for errors.	The program's code is responsible for exceptions.
5.	Error are defined in the java.lang.Error package.	Exception are defined in the java.lang.Exception package.